Tools and Technologies: Architecture & Tool Selection Rationale

Phase 2: Technology Stack Justification

Project: Global Olympic Sports Performance Analysis • **Author:** Md. Firdous Alam • **Era:** 1896–2016 Centennial Dataset

1. Technology Selection Rationale

Each component of the technology stack was selected to maximize analytical rigor, publication accessibility, and automation.

- Python & Pandas: Chosen over SQL/Excel for vectorized performance, programmatic testability, and reproducibility.
- Tableau Public: Chosen for enterprise-grade interactive dashboards accessible via standard web browsers without software installation.
- ReportLab 5.1.0: Selected for programmatic, pixel-accurate PDF document synthesis.
- Git & GitHub: Chosen for distributed version control, portfolio hosting, and open-source verification.